

COVID - 19
An Overview

DR. A. K. SHUKLA
M.D, FRCP (Edin), FICP, FICAM
Sr. Consultant Physician
Kailash Hospital & Heart Institute, Noida

In the year 2019, corona virus took the world by storm and caused a pandemic. Millions of people were killed. The one's living had fear and uncertainty in their eyes.

Initially all healthcare providers were unsure about the symptoms, diagnosis and prognosis of this virus. In spite of having their own lives at risk, healthcare providers across the globe, left no stone unturned in saving others. By the end of year 2020, covid 19 was declared a global health emergency by the WHO.

Covid 19 is believed to start as a mild disease with flu like symptoms, which eventually becomes severe with respiratory involvement leading to hospitalization and in most cases death.

The virus enters human cells by binding to angiotensin converting enzyme -2 (ACE-2) spike protein (S. Protein) in the lungs and transmits through respiratory droplets and aerosols from an infected person.

In the severe cases, an excessive immune activation is observed characterized by elevated inflammatory markers - Interleukin-6 (IL-6) C-Reactive Protein (CRP) and Ferritin, decreased lymphocytes and spectrum of end-organ damages, causing respiratory failure and multiorgan dysfunction. This is proceeded by high incidence of thrombosis (coagulation of blood in the vessels) resulting in critical illness and mortality.

The different waves of covid 19's spread, gave us an in-depth knowledge of its pathophysiology and how antiviral drugs like Remdesivir, Lopinavir, Ritonavir, Dexamethasone, Methylprednisolone along with Oxygen therapy could serve as a treatment option. Also, availability of drugs like Chloroquine and Ivermectin helped in treating the one's affected with the virus and its re-occurrence.

Various formulations of drugs like Tocilizumab were globally tested in efforts of containing the virus, but none of them proved effective.

As time progressed, hand hygiene specially the use of hand sanitizers, adorning PPE (masks, eyewear), rigorous contact tracing, isolation/quarantining in case of contact with the virus, emerged as the main stay factors preventing its spread.

After relentless efforts of research, speculation and in a very short time frame, molecular biologist and immunologists brought together vaccines which changed course for covid 19's path of human destruction.

COVID – 19 vaccination helps to make antibodies against the spike protein, this way the virus is blocked from entering the human cell and causing infection.

There are four main categories of vaccines: whole virus, protein subunit, viral vector, nucleic acids (RNA and DNA). The covid 19 vaccines available today are based on either one of the 4 categories:

1. Oxford Astra Zeneca Vaccines- **Covishield/ Covovax**
2. Bharat Biotech Vaccines- **Covaxin.**
3. Reddy's Lab- **Sputnik V Vaccine**
4. Biological E Ltd- **Carbevax**
5. Zydus Cadila- **ZyCoV –D**
6. Moderna – **Spikevax**
7. Pfizer- **Bio N tech vaccine**

Antibodies level peaks in roughly 3–5 weeks post vaccination. In people, who contacted the virus, this usually occurs after onset of symptoms or PCR diagnosis. As per government advisory, these vaccines were proven safe for pregnant woman and children.

But, today, the question in everyone's mind is, can this antibody level predict our probability of a re-infection and protection against it?

Currently, there are no evidences to predict the proficiency of natural immunity conferred by SARS - COV -2 antibodies against re-infection.

Even though we have seen mutations of covid 19 from SARS-COV-2, giving birth to Alfa Variant > Delta Variant and much recently the OMICRON (B.1.1.529), more new variants may be expected to come.

Thus, further research is needed for a better understanding of covid 19's diagnosis, treatment, prognosis and efforts for development of new improved vaccines and drugs for its prevention. So that we are together prepared and confident in handling what the futures holds.